

Backend Scope – LSP Platform

A comprehensive technical specification for building a secure, RBI-compliant lending service provider platform. This document outlines the complete backend architecture, APIs, third-party integrations, and compliance requirements for engineering teams developing a scalable digital lending solution.

Core Backend Architecture

The platform is built on a robust, cloud-ready infrastructure designed for scale and security. The architecture supports multiple client applications through RESTful APIs, ensuring seamless integration across mobile apps, web applications, and admin dashboards.

Security is embedded at every layer with JWT-based authentication, role-based access control, HTTPS enforcement, end-to-end encryption, and comprehensive audit logging. The system is designed to meet stringent RBI compliance requirements whilst maintaining optimal performance and scalability.

System Components Overview

API Layer

RESTful endpoints serving mobile, web, and admin clients with versioning support

Security Layer

JWT tokens, role-based access control, encryption, and session management

Compliance Engine

Audit logging, consent management, and RBI regulatory requirement enforcement

Integration Hub

Third-party service connectors for KYC, payments, e-sign, and credit bureaus

Authentication & User Management

The authentication system provides secure, passwordless login using OTP verification. It supports JWT token generation with refresh token mechanisms, device binding for added security, and optional biometric session management.

User profiles are managed centrally, allowing seamless updates whilst maintaining security through proper session handling and logout mechanisms. The system tracks device information and can enforce single-device policies when required.

- **Core Capabilities**

- OTP-based registration and login
- JWT token generation and refresh
- Device binding and biometrics
- Profile management
- Secure logout with token invalidation

Authentication API Endpoints



/auth/send-otp

Initiates OTP delivery to user's registered mobile number for authentication



/auth/verify-otp

Validates OTP and generates JWT access and refresh tokens



/auth/refresh-token

Issues new access token using valid refresh token



/auth/logout

Invalidates tokens and terminates active session



/user/profile

Retrieves and updates user profile information

KYC & Identity Verification Engine

The KYC engine provides comprehensive identity verification capabilities, integrating with multiple third-party services to validate PAN cards, Aadhaar documents, and bank accounts. The system performs OCR on uploaded documents and securely stores verification results.

All verification processes comply with RBI guidelines, maintaining audit trails and ensuring data privacy. The engine supports real-time verification through partner APIs whilst providing fallback mechanisms for manual review when automated verification fails.

KYC Integration Partners



Karza / Signzy / IDfy

PAN and Aadhaar verification services with real-time API integration



DigiLocker

Government-backed digital document repository for Aadhaar verification



Razorpay / Cashfree

Penny drop verification for bank account validation



Google ML Kit / AWS Textract

OCR engines for extracting text from document images

Verification Flow

Documents are uploaded through secure endpoints, processed via OCR, validated against authoritative databases, and stored with encrypted metadata. The system maintains complete audit trails of all verification attempts and results.

Failed verifications trigger manual review workflows whilst successful verifications automatically progress the application to the next stage. All sensitive documents are encrypted at rest with access controls enforced through RBAC.

KYC API Endpoints

1

/kyc/pan-verify

Validates PAN card details against government records

2

/kyc/aadhaar-verify

Verifies Aadhaar through DigiLocker integration

3

/kyc/bank-verify

Performs penny drop validation on bank account

4

/documents/upload

Securely uploads and processes KYC documents

Consent & Legal Compliance

The consent management system is critical for RBI compliance, maintaining explicit records of user consent for data processing, credit bureau checks, and information sharing with lending partners. Every consent action is timestamped and stored with complete audit trails.

Users can view their consent history at any time and revoke permissions where applicable. The system maintains versioned legal documents (terms of service, privacy policies) and ensures users acknowledge the current versions before proceeding with loan applications.

Consent Management Capabilities

Compliance Requirements

The consent engine enforces RBI guidelines by recording explicit user agreement before accessing credit information or sharing data with NBFCs. All consent actions include purpose specification, scope limitation, and user acknowledgement.

The system maintains tamper-proof audit logs of consent grants and revocations, supporting regulatory audits and dispute resolution. Legal document versions are tracked, ensuring users always agree to current terms.

Consent Recording

Capture and store explicit user consent with metadata

History Tracking

Maintain complete timeline of consent actions

Revocation Support

Allow users to withdraw consent and enforce data handling

Legal Compliance API Endpoints

/consent/record

Records new user consent with type, purpose, and timestamp

/consent/history

Retrieves complete consent timeline for a user

/consent/revoke

Processes consent revocation and triggers data handling policies

/legal/terms

Returns current terms of service version

/legal/privacy-policy

Provides current privacy policy document

Loan Eligibility & Credit Engine

The credit engine evaluates loan eligibility based on income verification, employment details, existing obligations, and soft credit bureau pulls. It calculates maximum loan amounts, applicable interest rates, and EMI structures without impacting the user's credit score through hard inquiries.

The engine compares offerings from multiple NBFC partners, presenting users with the most suitable options based on their credit profile. All calculations follow RBI guidelines for transparency in lending, clearly displaying APR, processing fees, and total repayment amounts.

Credit Engine Components

01

Eligibility Assessment

Evaluates income, employment stability, and existing liabilities to determine loan qualification

03

EMI Calculation

Computes monthly instalments, interest rates, and total repayment amounts

02

Credit Bureau Check

Performs soft inquiry to access credit history without impacting credit score

04

Lender Matching

Compares NBFC offerings and recommends best options based on user profile

Credit Engine API Endpoints

/eligibility/check

Determines loan eligibility and maximum loan amount based on user profile

/lenders/compare

Returns comparison of available lenders with rates and terms

/loan/calculate

Computes EMI schedules, interest amounts, and repayment terms

/loan/select-lender

Records user's lender selection and initiates application process

Loan Application Processing System

The application processing system manages the complete loan application lifecycle through a multi-step workflow. Applications are created, validated, enriched with reference verification data, and submitted to partner NBFCs through secure API integrations.

Each application receives a unique reference number for tracking. The system maintains application state across sessions, allowing users to save progress and resume later. All submitted applications undergo automated validation checks before forwarding to lending partners.

Application Processing Features



Multi-Step Forms

Progressive data collection with validation at each stage and session persistence



Reference Verification

Captures and validates personal and professional references as required by lenders



Status Tracking

Maintains complete lifecycle state from draft to submission to closure



NBFC Submission

Securely transmits application data to partner NBFCs through encrypted channels



Reference Generation

Creates unique application identifiers for tracking and customer support

Application API Endpoints



/loan/application

Creates new application or updates existing draft with provided data



/loan/submit

Validates and submits completed application to selected NBFC partner



/loan/application/{id}

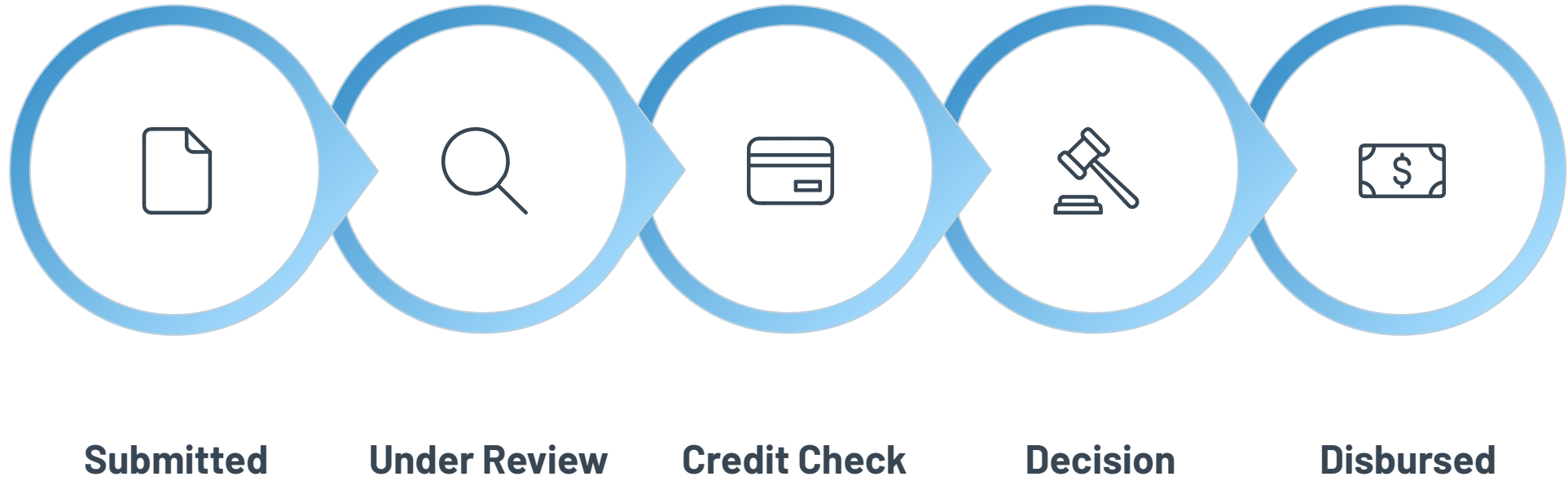
Retrieves complete application details including current status and history

Application Tracking & Workflow Engine

The workflow engine provides real-time status updates throughout the loan application journey. It processes webhooks from NBFC partners, updates application states, and triggers notifications to keep users informed of progress.

The system maintains detailed status timelines showing when each milestone was reached, including submission timestamps, credit check completion, approval decisions, and disbursement dates. This transparency builds user trust and reduces support queries.

Application Lifecycle Statuses



Applications progress through clearly defined stages from submission to closure. Each status change is logged with timestamps and triggering events. Users receive notifications at key milestones, and the admin dashboard provides visibility into applications stuck at any stage for proactive intervention.

Workflow Engine Capabilities

Real-Time Updates

The engine processes status updates from multiple sources including manual admin actions, automated system events, and webhook callbacks from lending partners. All state changes are atomic and maintain consistency across distributed systems.

Timeline views show complete application history with timestamps, status transitions, and associated events. This audit trail supports dispute resolution and provides transparency for regulatory compliance.

Status Management

Real-time state tracking with validation rules

Webhook Processing

Secure handling of partner callbacks

Notification Triggers

Event-based alerts via push, SMS, email

Tracking API Endpoints



/loan/applications

Lists all applications for user with current status



/loan/application/{id}/status

Returns current status with last update timestamp



/loan/application/{id}/timeline

Provides complete status history with transitions

E-Agreement & E-Sign System

The e-sign system generates loan agreements dynamically based on approved terms and facilitates legally binding electronic signatures using Aadhaar-based authentication. All agreements are generated from templates with borrower-specific details populated automatically.

Signed documents are stored securely with cryptographic proofs of signing, timestamps, and signer authentication details. The system integrates with NSDL, DigiLocker eSign, and eMudhra to provide multiple signing options whilst ensuring legal compliance and enforceability.

E-Sign Integration & APIs



Agreement Generation

Dynamic PDF creation from templates with borrower and loan details



Aadhaar E-Sign

OTP-based signing with Aadhaar authentication through partner APIs



Secure Storage

Encrypted document storage with access controls and audit trails



Verification

Cryptographic validation of signatures and document integrity

API Endpoints

/loan/agreement/{id}

Generates and retrieves loan agreement PDF

/loan/esign/initiate

Starts e-sign workflow with partner service

/loan/esign/verify

Validates signature and updates loan status

Disbursement & Payment Tracking

The disbursement module tracks when approved loans are transferred from NBFC partners to borrower bank accounts. It maintains records of UTR numbers, transaction timestamps, and disbursement amounts, reconciling them against approved loan terms.

Payment tracking provides real-time visibility into outstanding balances, upcoming EMIs, and repayment history. The system integrates with payment gateways to record manual payments whilst also processing automatic debit mandates for EMI collection.

Disbursement & Payment APIs



/loan/disbursement/{id}/status

Checks and updates disbursement status with UTR tracking



/payment/record

Records manual or automated payments against loan accounts



/loan/{id}/outstanding

Calculates current outstanding balance including interest and penalties

These endpoints ensure complete financial tracking and reconciliation, supporting accurate accounting and customer service operations.